THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **Exploring the Investigative World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): A fair test changes only one factor at a time. Reason (R): Changing several factors together makes it impossible to know which caused the effect. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A student studies the cooling of tea. Temperatures at 0, 2, 4, 6 minutes are 90, 78, 69, 62 degrees C. The fall during the first 2 minutes compared with the last 2 minutes is: 1

(a) smaller **(b)** equal **(c)** zero **(d)** greater

3. A thermometer reads 3 degrees C in melting ice. A reading of 53 degrees C for warm water corresponds to a true value of: 1

(a) 53 degrees C **(b)** 50 degrees C **(c)** 56 degrees C **(d)** 47 degrees C

4. Assertion (A): An experiment should be repeated several times. Reason (R): Repetition ensures that the hypothesis is always true. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Seeds in cups: A (dry cotton), B (wet cotton, kept in light), C (wet cotton, kept in dark). Which pair isolates the effect of light? 1

(a) B and C **(b)** A, B and C together
(c) A and B **(d)** A and C

6. Which of these shows a good use of a control in a test on a new fertiliser? 1

(a) all plants given fertiliser
(b) different plants different soil
(c) no plants
(d) plants without fertiliser grown in same conditions

7. Assertion (A): A hypothesis can be rejected by an experiment. Reason (R): A hypothesis must be a testable statement. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A student tests four detergents but uses different amounts of water for each. The flaw is: 1

(a) wrong cloth colour **(b)** water amount is not controlled
(c) stain is a quality **(d)** too many detergents

9. **Assertion (A):** In a fair test on how oil temperature affects the puffing of a puri, the thickness of the dough must be kept the same. 1

Reason (R): Only one variable should be changed at a time while all other factors are kept constant.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** A hypothesis can be tested by an experiment or observation. 1

Reason (R): A hypothesis is a tentative, testable explanation or prediction.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A student's pendulum gives 20 swings in 34 s. Find the time period and explain why 20 swings are timed. 2

12. Give two differences between accuracy and precision. 2

13. Why are tables with units preferred over writing results in sentences? 2

14. The mean of five readings is 12.4 s. Four readings are 12.0, 12.6, 12.5 and 12.3. Find the fifth. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Critically evaluate: 'We tested only one type of soil, so all plants grow best in garden soil.' 3

16. From a graph, how do you judge whether two quantities are directly proportional? 3

17. Suggest how you will test the claim that a particular kind of paper towel absorbs more water than another. 3

OR

A student records 5 readings of the time taken for a ball to roll: 4.1, 4.0, 4.2, 6.5, 4.1 s. Identify the outlier, suggest its cause and give the best mean. 3

18. Explain how bias can affect an investigation and how to reduce it. 3

OR

Differentiate between an experiment and a survey with an example each. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Meera wants to know whether the amount of water in the dough changes how well a puri puffs. She makes four puris using 20 mL, 30 mL, 40 mL and 50 mL of water for the same quantity of flour, rolls them to equal thickness and fries them in oil at 180 degrees C. She measures the time each takes to puff. 5

(i) Name the independent variable.

(ii) Name the dependent variable.

(iii) Name one variable that Meera has kept constant.

(iv) Write a testable hypothesis for this study and explain why she should repeat the trial.

20. A student's results for plant height (cm) after 4 weeks with watering of 0, 20, 40, 60, 80 mL were 3, 8, 12, 13, 9. Describe the pattern, suggest an explanation and a next experiment. 5

OR

Heights of ten students in cm: 140, 142, 145, 150, 150, 151, 153, 155, 156, 158. Find the mean and the range and say which graph suits the data. 5

END OF QUESTION PAPER — Total: 40 Marks